UNIVERSITY OF ZAMBIA

CSC 6302 — Computer Vision

Practical 3 Report: Detecting Things

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Media: consented group photo (*images/group_faces.jpg*); phone street clip (*images/street_clip.mp4*); everyday object (*images/object.jpg*).

What I did is walk through template matching, Haar face detection, YOLOv8n, and a Gradio Haar/YOLO app. Composer helped with script layout; parameter choices, timing, and the ethics section are mine.

Task 1 — Template matching

I cropped a patch from my object photo and used **cv2.matchTemplate** (TM_CCOEFF_NORMED). Same-view score is 1.00. At 0.55× scale the score drops (~0.77) and the box drifts; after a 25° rotation the peak is ~0.66. A fixed template needs matching scale and orientation — that is why cascades and YOLO exist.

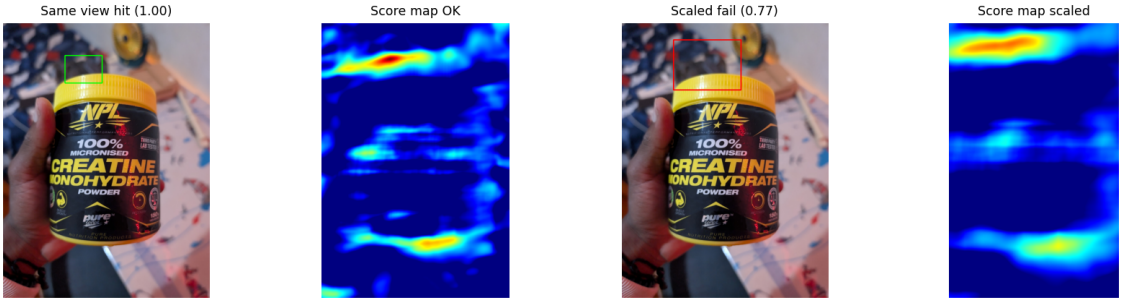


Figure 1. Same-view hit vs scaled failure.

Task 2 — Haar / Viola–Jones faces

On my group photo I swept scaleFactor ∈ {1.05, 1.1, 1.3} and minNeighbors ∈ {3, 5, 8}. Looser settings keep more faces; 1.3 / 8 starts missing people. Default 1.1 / 5 is a sensible middle. I also annotated every 3rd frame of my phone street clip (100 frames).



Figure 2. Haar on the group photo.

scaleFactor	minNeighbors	detections	hits	misses	false_positives
1.05	3	3	3	1	0
1.05	5	1	1	3	0
1.05	8	0	0	4	0
1.1	3	2	2	2	0
1.1	5	1	1	3	0
1.1	8	0	0	4	0
1.3	3	0	0	4	0
1.3	5	0	0	4	0
1.3	8	0	0	4	0

Table 1. Haar parameter study.

Task 3 — YOLO object detection

YOLOv8n on group / street still / object shows the confidence trade-off (group 7→5 boxes from conf 0.1→0.75; street 6→1). Loose vs strict NMS shows why overlapping duplicates get suppressed.

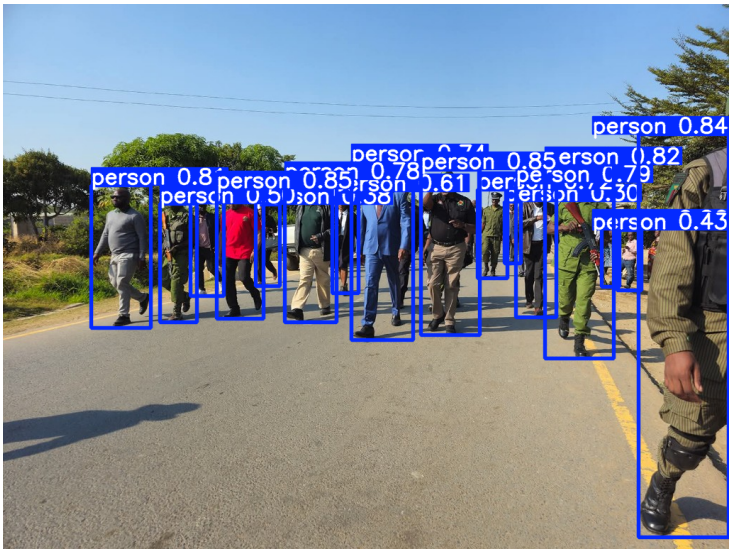


Figure 3. YOLO on the group photo (conf=0.25).



Figure 4. YOLO on a street-clip frame.



Figure 5. NMS / confidence comparison.

Timing on 25 resized street frames: Haar ≈ 243 ms/frame, YOLO ≈ 149 ms/frame on this laptop. Haar is face-only; YOLO covers many classes and handles pose better.

Task 4 — Gradio app + Hugging Face Spaces

App: *prac3/app.py* — upload an image, choose Haar or YOLO, move the confidence / strictness slider. Local: *python prac3/app.py*. HF Spaces hosts the same Gradio app online (upload app.py + opencv-python-headless, ultralytics, gradio, numpy). Suggested Space name: **csc6302-prac3-SINYANI**.

Task 5 — Ethics

I only used a group photo people agreed I could process for coursework. I will not put identifiable classmate faces on a public Space without blur/consent. Street clip is my own phone video of a public road; I still avoid posting frames that single people out. Detectors can false-positive — any real counting use needs a human in the loop.

References

Bradski, G. & Kaehler, A. — Learning OpenCV.

Viola, P. & Jones, M. (2001). Rapid object detection using a boosted cascade of simple features.

Ultralytics YOLOv8 documentation.

OpenCV docs: CascadeClassifier, matchTemplate.

AI disclosure: Cursor Composer helped with script layout and Gradio wiring. I ran the pipelines and wrote this report.